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T5.1 解：式(5.67)：$^0J(\theta)=\begin{pmatrix} -l_1s_1-l_2s_{12} & -l_2s_{12} \\ l_1c_1+l_2c_{12} & l_2c_{12} \end{pmatrix}$.

由例5.4，$DET[^0J(\theta)]=(-l_1s_1-l_2s_{12})(l_2c_{12})+l_2s_{12}(l_1c_1+l_2c_{12})$

$=l_1l_2s_2$

令 $l_1l_2s_2=0$ $\Rightarrow\theta$

$\Rightarrow\theta_2$ 为 0° 或 180° 时处于奇异位形，与例5.3结果一致

T5.3 解：$P_B=\underbrace{{}^B_AR\cdot P_A}$，即是 $P_B={}^B_AR\,P_A$. 从下标坐标系到上标坐标系.

① 从基座到末端的速度传递. ← 表示坐标系B相对于A的旋转

式(5.12)：$^AV_Q={}^A_BR(^BV_Q)+{}^A\Omega_B\times{}^A_BR\,{}^BQ$

式(5.45)

$$^{i+1}w_{i+1}={}^{i+1}_iR\,{}^iw_i+\dot\theta_{i+1}\,{}^{i+1}\hat Z_{i+1}$$

式(5.47)

$$^{i+1}V_{i+1}={}^{i+1}_iR(^iV_i+{}^iw_i\times{}^iP_{i+1}).$$

~~② 计 w_{i+1} 与 i~~

已知 $^0w_0=0$，$^0V_0=0$.

$$\Rightarrow\begin{cases} ^1w_1={}^1_0R\,{}^0w_0+\dot\theta_1\,{}^1\hat Z_1=\dot\theta_1\,{}^1\hat Z_1 \\ ^1V_1={}^1_0R(^0V_0+{}^0w_0\times{}^0P_1)=0 \end{cases}$$

$$\Rightarrow\begin{cases} ^2w_2=\begin{bmatrix} s_2\dot\theta_1 \\ c_2\dot\theta_1 \end{bmatrix} \\ ^2V_2=\begin{bmatrix} 0 \\ -l_1\dot\theta_1 \end{bmatrix}\dot\theta_2 \end{cases}$$ 重复运用公式.

$$\Rightarrow\begin{cases} ^4w_4={}^4_3R\,{}^3w_3+\dot\theta_4\,{}^4\hat Z_4=\begin{bmatrix} s_{23}\dot\theta_1 \\ c_{23}\dot\theta_1 \end{bmatrix}\dot\theta_2+\dot\theta_3 \\ ^4V_4={}^4_3R\,(^3V_3+{}^3w_3\times{}^3P_4)=\begin{bmatrix} l_2s_3\dot\theta_2 \\ l_2c_3\dot\theta_2+l_3(\dot\theta_2+\dot\theta_3) \\ -l_1\dot\theta_1-l_2c_2\dot\theta_1-l_3c_{23}\dot\theta_1 \end{bmatrix} \end{cases}$$

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: ${}^4J(\theta)$ 由 ${}^4V_4 = {}^4J(\theta)\cdot\dot{\theta}$

得 ${}^4J(\theta) = \begin{bmatrix} 0 & l_2 s_3 & 0 \\ 0 & l_2 c_3 + l_3 & l_3 \\ -l_1 - l_2 c_2 - l_3 c_{23} & 0 & 0 \end{bmatrix}$

② 从末端到基座的静力传递.

$$\begin{cases} {}^i f_i = {}^i_{i+1}R\ {}^{i+1}f_{i+1} \\ {}^i n_i = {}^i_{i+1}R\ {}^{i+1}n_{i+1} + {}^i P_{i+1} \times {}^i f_i \end{cases}$$

$$\Rightarrow \begin{cases} {}^3 f_3 = {}^3_4 R\ {}^4 f_4 = \begin{bmatrix} f_x \\ f_y \\ f_z \end{bmatrix} \\ {}^3 n_3 = {}^3_4 R\ {}^4 n_4 + {}^3 P_4 \times {}^3 f_3 = \begin{bmatrix} 0 \\ -l_3 f_z \\ l_3 f_y \end{bmatrix} \end{cases}$$

$$\Rightarrow \begin{cases} {}^2 f_2 = \begin{bmatrix} c_3 f_x - s_3 f_y \\ s_3 f_x + c_3 f_y \\ f_z \end{bmatrix} \\ {}^2 n_2 = \begin{bmatrix} l_3 s_3 f_z \\ -l_2 f_z - l_3 c_3 f_z \\ -l_2(s_3 f_x + c_3 f_y) + l_3 f_y \end{bmatrix} \end{cases}$$

$$\Rightarrow \begin{cases} {}^1 f_1 = {}^1_2 R\, f_2 \\ {}^1 n_1 = {}^1_2 R\, {}^2 n_2 + {}^1 P_2 \times {}^1 f_1 \end{cases}$$

又有 $\tau_i = {}^i n_i{}^T\, {}^i\hat{z}_i$

$$\Rightarrow \quad \tau_1 = [-l_1 - l_2 c_2 + l_3(s_2 s_3 - c_2 c_3)] f_z$$

$$\tau_2 = l_2 s_3 f_x + (l_2 c_3 + l_3) f_y$$

$$\tau_3 = l_3 f_y$$

由 $\tau = {}^4J^T(\theta) \begin{bmatrix} f_x \\ f_y \\ f_z \end{bmatrix}$

$$\Rightarrow \quad {}^4J(\theta) = \begin{bmatrix} 0 & l_2 s_3 & 0 \\ 0 & l_2 c_3 + l_3 & l_3 \\ -l_1 - l_2 c_2 - l_3 c_{23} & 0 & 0 \end{bmatrix}.$$

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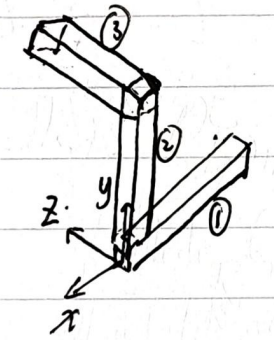
③直接对运动学方程微分

$${}^0P_{4ORG} = \begin{bmatrix} l_1c_1 + l_2c_1c_2 + l_3c_1c_{23} \\ l_1s_1 + l_2s_1c_2 + l_3s_1c_{23} \\ l_2s_2 + l_3s_{23} \end{bmatrix}$$

$${}^0J(\theta) = \frac{\partial P}{\partial \theta} = \begin{bmatrix} -l_1s_1 - l_2s_1c_2 - l_3s_1c_{23} & -l_2c_1s_2 - l_3c_1s_{23} & -l_3c_1s_{23} \\ l_1c_1 + l_2c_1c_2 + l_3c_1c_{23} & l_2s_1s_2 - l_3s_1s_{23} & l_3s_1s_{23} \\ 0 & l_2c_2 + l_3c_{23} & l_3c_{23} \end{bmatrix}$$

$$\Rightarrow {}^4J(\theta) = {}^4_0R \cdot {}^0J(\theta) = \begin{bmatrix} 0 & l_2s_3 & 0 \\ 0 & l_2c_3 + l_3 & l_3 \\ -l_1 - l_2c_2 - l_3c_{23} & 0 & 0 \end{bmatrix}$$

T5.7 解：已知 $J(\theta) = \begin{bmatrix} 1 & , & , \end{bmatrix}$，考虑3个互相垂直的平移关节，首尾相连。



如左图，平移关节①②③分别沿x、y、z轴方向。

运动学描述：笛卡尔：由3个相互垂直的平移关节首尾相连得到的3自由度机构。每个关节的位移、速度加速度即末端执行器的位移、速度、加速度。

T5.10

解：

$$\tau = {}^3J^T \, {}^3F$$

$$\Rightarrow {}^3F = ({}^3J^T)^{-1}\tau \quad , \text{已知 } {}^3J(\theta) = \begin{bmatrix} l_1s_2 & 0 \\ l_1c_2 + l_2 & l_2 \end{bmatrix}$$

$$\Rightarrow \text{变换矩阵 } ({}^3J^T)^{-1} = \frac{1}{l_1l_2s_2} \begin{bmatrix} l_2 & -l_1c_2 - l_2 \\ 0 & l_1s_2 \end{bmatrix}$$

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T5.12 解: 已知 ${}^4J(\theta) = \begin{bmatrix} 0 & l_2 s_3 & 0 \\ 0 & l_2 c_3 + l_3 & l_3 \\ -l_1 - l_2 c_2 - l_3 c_{23} & 0 & 0 \end{bmatrix}$

$DET[{}^4J(\theta)] = -l_2 l_3 s_3 (l_1 + l_2 c_2 + l_3 c_{23})$

令 $DET[{}^4J(\theta)] = 0$. 奇异位形的.

=> ① 在工作空间边界的关节角 $\{\theta_1, \theta_2, 0\}$.

② 在一内部的关节角 $\{\theta_1, \theta_2, \theta_3\}$

满足. $l_1 + l_2 c_2 + l_3 c_{23} = 0$

即 $c_2 \cdot (l_2 + l_3 c_3) = -l_1$

$c_2 = \frac{-l_1}{l_2 + l_3 c_3}$.

T5.13 解. $\tau = [{}^0J(\theta)]^T {}^0F = \begin{bmatrix} -l_0(l_1 s_1 + l_2 s_{12}) \\ -l_0 s_{12} - l_2 \end{bmatrix}$

即 $\begin{cases} \tau_1 = -l_0(l_1 s_1 + l_2 s_{12}) \\ \tau_2 = -l_0 l_2 s_{12} \end{cases}$

T5.18 解: ${}^0P_{3ORG} = \begin{bmatrix} l_1 c_1 + l_2 c_1 c_2 \\ l_1 s_1 + l_2 s_1 s_2 \\ l_2 s_2 \end{bmatrix}$

$\Rightarrow {}^0J(\theta) = \frac{\partial P}{\partial \theta} = \begin{bmatrix} -l_1 s_1 - l_2 s_1 c_2 & -l_2 c_1 s_2 & 0 \\ l_1 c_1 + l_2 c_1 c_2 & -l_2 s_1 s_2 & 0 \\ 0 & l_2 c_2 & 0 \end{bmatrix}$.

$\Rightarrow {}^0V_3 = {}^0J(\theta) \cdot \dot{\theta}$

$= \begin{bmatrix} -(l_1 s_1 + l_2 s_1 c_2)\dot{\theta}_1 - l_2 c_1 s_2 \dot{\theta}_2 \\ (l_1 c_1 + l_2 c_1 c_2)\dot{\theta}_1 - l_2 s_1 s_2 \dot{\theta}_2 \\ l_2 c_2 \dot{\theta}_2 \end{bmatrix}$

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